

BERNARDO NANDANIAR SUNIA

NLP Engineer • Data Scientist • Machine Learning Engineer

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PROFESSIONAL SUMMARY

Informatics graduate from Diponegoro University (GPA 3.78, Cum Laude) with hands-on experience building NLP systems end to end, from labeling and modeling through deployment. Comfortable working across chatbots, sentiment analysis, and multilingual semantic search using PyTorch and transformer models such as BERT, SBERT, and mBERT. Especially strong in Indonesian-language text, where sarcasm, negation, code-mixing, and informal phrasing tend to trip up simpler models.

TECHNICAL SKILLS

Programming & Core Tools: Python, SQL, Git, Docker
NLP & Machine Learning: PyTorch, BERT / SBERT / mBERT, Sentence-Transformers, Scikit-learn, FAISS, NLTK, Sastrawi
Data Analysis & Visualization: Pandas, NumPy, SciPy, Statsmodels, Matplotlib, Plotly, Streamlit, Jupyter
Methods: NLP, Machine Learning, Information Retrieval, Requirements Gathering, QA Testing, Statistical Modeling
Languages: Indonesian (Native), English (CEFR 488)

PROFESSIONAL EXPERIENCE

- IT Analyst (Full-Time)** May 2026 – Jul 2026
PT Wiraky Nusa Telekomunikasi, Bandung, Indonesia
 - Designed and fine-tuned an Aspect Sentiment Quad Extraction (ASQE) model to pull out aspects, opinions, categories, and sentiment from over 5,000 self-labeled reviews, paying particular attention to sarcasm, negation, code-mixing, and informal Indonesian phrasing.
 - Built and deployed an NLP-powered chatbot that answers queries through embedding-based similarity matching, then added a CRUD admin dashboard and automated data retention so the knowledge base stays easy to manage.
 - Engineered a secure review-analytics platform that combines multi-signal anomaly detection with interactive dashboards, plus a dedicated module for property-investment analytics.
- Chatbot Developer Intern, Automated Information System** May 2025 – Jul 2025
Class II Ambarawa Correctional Facility, Ambarawa, Indonesia
 - Led the chatbot development process from start to finish, running stakeholder interviews, defining requirements, optimizing the training dataset, and handling QA and post-launch monitoring.
- Laboratory Teaching Assistant** Jun 2023 – Jun 2025
Computer Laboratory, Diponegoro University, Semarang, Indonesia
 - Mentored 20+ students in Python and algorithmic problem-solving, and designed structured competency assessments that helped raise average lab grades by 15%.
- Financial Reporting Application Developer Intern** Dec 2024 – Feb 2025
East Semarang District, Semarang, Indonesia
 - Developed a full-stack financial-reporting application with ReactJS, JavaScript, and SQL, handling everything from requirements gathering to UI development and staff training, which improved data accuracy and day-to-day efficiency.
- Assistant Manager, Supply Chain & Creative Design (Part-Time)** Jan 2022 – Dec 2024
YC Electric, Semarang, Indonesia
 - Optimized inventory tracking and supplier logistics, and produced 30+ promotional materials using digital design tools that helped strengthen brand visibility and marketing reach.

KEY PROJECTS

- Aspect Sentiment Classification: TF-IDF vs. SBERT on Shopee Reviews** Aug 2025 – Feb 2026
Tech Stack: Jupyter Notebook, Scikit-learn, Sentence-Transformers, Pandas, SciPy, Sastrawi, Statsmodels
 - Curated a hand-labeled dataset of 1,000+ Shopee reviews and compared TF-IDF against SBERT representations paired with Logistic Regression; SBERT came out ahead with a 94% F1-score.
- Custom Search Engine Using VSM and LSI** Mar 2024 – May 2024
Tech Stack: Python, Jupyter Notebook, Streamlit, Pandas, NumPy, Sastrawi, NLTK, Scikit-learn, Matplotlib
 - Created an interactive document-search engine on the Vector Space Model and Latent Semantic Indexing, using text preprocessing, TF-IDF, cosine similarity, and SVD-based semantic analysis to improve retrieval quality.
- Advanced Information Retrieval System Using ML & Deep Learning** Aug 2024 – Dec 2024
Tech Stack: Jupyter Notebook, Streamlit, Pandas, NumPy, mBERT, PyTorch, Matplotlib, Plotly
 - Implemented a multilingual semantic-search system on a fine-tuned mBERT paired with FAISS, allowing fast, accurate document retrieval through dense embedding similarity search.

EDUCATION

Bachelor of Informatics, Diponegoro University, Semarang, Indonesia (GPA 3.78, Cum Laude) Aug 2022 – Mar 2026

CERTIFICATIONS

- Intellectual Property Certificate – Software, Ministry of Law (2026)
- Intellectual Property Certificate – Chatbot, Ministry of Law (2025)
- Data Classification and Summarization, IBM SkillsBuild (2025)
- PyTorch & Generative AI, Avalon AI (2024)
- SQL & Java Programming, Oracle Academy (2024)
- Cloud Computing, Alibaba Cloud (2024)
- Data Analysis Bootcamp, Diponegoro University (2023)